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Mechanical modeling of Thin Films Bonded to Functionally Graded Materials

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Abstract. In this study the contact problems of thin films bonded to Functionally Graded Materials (FGM) are considered. In these problems the loading consists of any one or combination of stresses caused by uniform temperature changes and temperature excursions, far field mechanical loading, and residual stresses resulting from film processing or in the manufacturing process of the graded coating. The primary interest in this study is in examining stress concentrations or singularities near the film ends for the purpose of addressing the question of crack initiation and propagation in the substrate or along the interface. The underlying contact mechanics problem is formulated by assuming that the film is a "membrane" and the FGM an elastic continuum, and is solved analytically by reducing it to an integral equation. The calculated results are the interfacial shear stress between the film and the graded substrate, mode II stress intensity factor at the end of the film and the axial normal stress in the film.

Introduction

Engineers try to produce smarter structures (systems) by utilizing nanotechnology and smart production methods through mimicking the perfect structures in the nature. For these structures to be more energy and cost efficient, and to have better performance while being ecologically compatible, they should be produced on smaller scales. Thin films or coated structures are one of these type of structures which are used with increasing demand in microelectronic integrated circuits, magnetic data storage systems, optical coatings (filters), abrasion/corrosion resistant coatings, solar cells, fuel cells and force transducers in micro-electro-mechanical systems[1-2].

There can be severe stresses developed during or after the production stage of thin films. This is a technological barrier – even if load carrying is not the main functional characteristic of the material [3]. Produced thin films should have enough strength to sustain these stresses. Therefore, structures with thin films should be mechanically modeled and attained analytical/numerical results should be used to improve production techniques and to define usage conditions.

Film stresses can be divided into two main categories: intrinsic and extrinsic. Intrinsic stresses are created during the development of the film on a substrate or on a neighborhood layer; extrinsic stresses, on the other hand, are the stresses caused by environmental conditions after the development period. To cope with intrinsic stresses, smart production techniques are developed [3,4].

In this study the contact problem for thin films bonded to graded coatings is considered. The loading in these type of problems, generally consists of any one or combination of stresses caused by uniform temperature changes and temperature excursions, far field mechanical loading, and residual stresses resulting from film processing or in the manufacturing process of the graded coating. The primary interest in this study is in examining stress concentrations or singularities near the film ends for the purpose of addressing the question of crack initiation and propagation in the substrate or along the interface. At the ends of the film, there are integrable contact shear stress singularities where the normal stress gradients in the film are the maximum, [5,6].

The problem of finite length film (e.g. an elastic stiffener) bonded to a half space substrate has been studied since 1930's and in the course of time, new solution techniques have been developed to solve the integro-differential or integral equation resulted from the formulation of the mixed boundary value problem [7,8]. There are many studies on singular stress field at the free edge that causes crack formation [7–18]. In this work, the numerical solution technique proposed by Erdogan and Gupta [6,7] is preferred due to its simplicity and effectiveness.

In this study, tri-material system is modeled as an isotropic continuum by using linear plane elasticity theory. In the model, a homogeneous film is assumed to be bonded to a functionally graded coating on a homogeneous substrate. Perfect bonding is assumed, i.e. displacement continuity prevails along the interfaces. The film is assumed to have a negligible thickness when compared to the other dimensions of the problem and hence modeled as a membrane. The loading on the system is taken as a constant strain, , distributed uniformly in the coating and substrate (see Fig. 1).

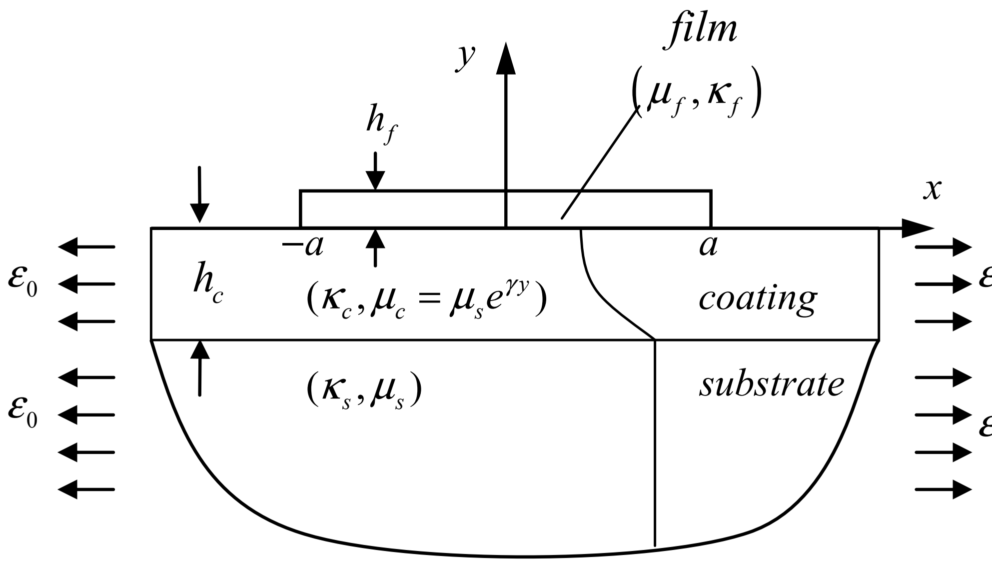

Figure 1. Geometry of the problem

Formulation of the Plane Elasticity Problem

Consider the plane elasticity problem shown in Figure 1 where the thickness of the film and the functionally graded coating are denoted by h_f and h_c respectively. The subscripts or subscripts f, c and s given in the formulation refer to film, coating and substrate respectively. The shear moduli μ_s, μ_f is constant and $\mu_c(y)$ is approximated by

$$\mu_c(y) = \mu_0 e^{\gamma y}, \qquad -h_c < y < 0, \tag{1}$$

where γ is a constant characterizing the material inhomogeneity, μ_0 is the value of $\mu_c(y)$ at the surface of the graded coating.

In the graded medium $-h_c < y < 0$ the spatial variation of the Poisson's ratio is assumed to be negligible.

The strain in the FGM coating can be written as

$$\frac{\partial}{\partial x} u_c(x,0) = \frac{\kappa_c + 1}{4\mu_2} \frac{1}{\pi} \int_{-\infty}^{\infty} \left[\frac{1}{t-x} + k_{21}(t,x) \right] \tau_{xy}^f(t,0)\, dt. \tag{2}$$

Compatibility condition, $\varepsilon_{xx}^f(x,0) = \varepsilon_{xx}^c(x,0)$, gives the integral equation of the problem:

$$\frac{1}{\pi}\int_{-a}^{a}\left[\frac{1}{t-x}-k_{21}(t,x)\right]\tau_{xy}^{f}(t,0)\,dt-\frac{\lambda}{2h_f}\int_{-a}^{x}\tau_{xy}^{f}(t,0)\,dt=-\frac{4\mu_0}{\kappa_c+1}\varepsilon_0, \tag{3}$$

where

$$\lambda=\frac{1+\kappa_f}{1+\kappa_c}\frac{\mu_0}{\mu_f}, \tag{4}$$

and $k_{21}(t,x)$ is a bounded function given in [19].

Solution of the Integral Equation

By defining the following normalized quantities:

$$x=ar,\quad t=as,\quad -a<(x,t)<a,\quad -1<(r,s)<1, \tag{5a,c}$$

$$\tau_{xy}^{f}(x,0)=p_1 p(r),\quad p_1=-\frac{4\mu_0}{\kappa_c+1}\varepsilon_0, \tag{5d,e}$$

and assuming a solution of the form:

$$p(s)=\phi(s)\frac{1}{\sqrt{1-s^2}},\quad \phi(s)=\sum_{0}^{\infty}a_n T_n(s), \tag{6a,b}$$

and using the orthogonality conditions and properties of the Chebyshev polynomials, it may be seen that $a_0=0$, and the integral equation (4) may be reduced to

$$\sum_{1}^{\infty}a_n\left[U_{n-1}(r)+R_n(r)\right]=1,\qquad -1<r<1 \tag{7}$$

where

$$R_n(r)=-\frac{1}{\pi}\int_{-1}^{1}\frac{T_n(s)}{\sqrt{1-s^2}}K_{21}(s,r)\,ds+\frac{\lambda}{2n}\frac{a}{h_f}U_{n-1}(r)\sqrt{1-r^2}. \tag{8}$$

Results and Discussions

In the problem considered, there are four main variables in the model of which two are mechanical type and the other two are geometrical type. The mechanical parameters are , stiffness ratio of the graded coating, $\Gamma=\mu_s/\mu_0$ and stiffness ratio of the coating upper surface to the film, $\lambda=\mu_0/\mu_f$. The geometrical parameters are aspect ratio of the film, l/h_f, and the film length to coating thickness ratio, l/h_c respectively . Note that the parameter Γ shows the grading ratio of the coating, and lambda shows the compliancy of the film with respect to the coating upper surface. The full parametric study of the problem seems to be practically impossible. One may, however, select two of the variables as main parameters and compute and plot the field quantities that deem to be important. The values of the parameters are selected so as to cover hitherto used values in the literature for the similar models [12, 20-27] and to be inside the domain as far as the efficiency of the numerical technique allows [12].

In Fig. 2, the variation of the normal stress, $\sigma_{xx}^{f}(x)$, in the film and the shear stress, $\tau_{xy}^{f}(x,0)$, on the film-coating interface is shown with respect to the various values of the parameters Γ. Finally, for different values of the parameters, variation in the mode II stress intensity factor, k_2, at the film ends is given in the Fig. 3. Graphs in the figures are shown only in one quadrant due to the symmetric variation of the resulting stresses on account of the symmetric loading.

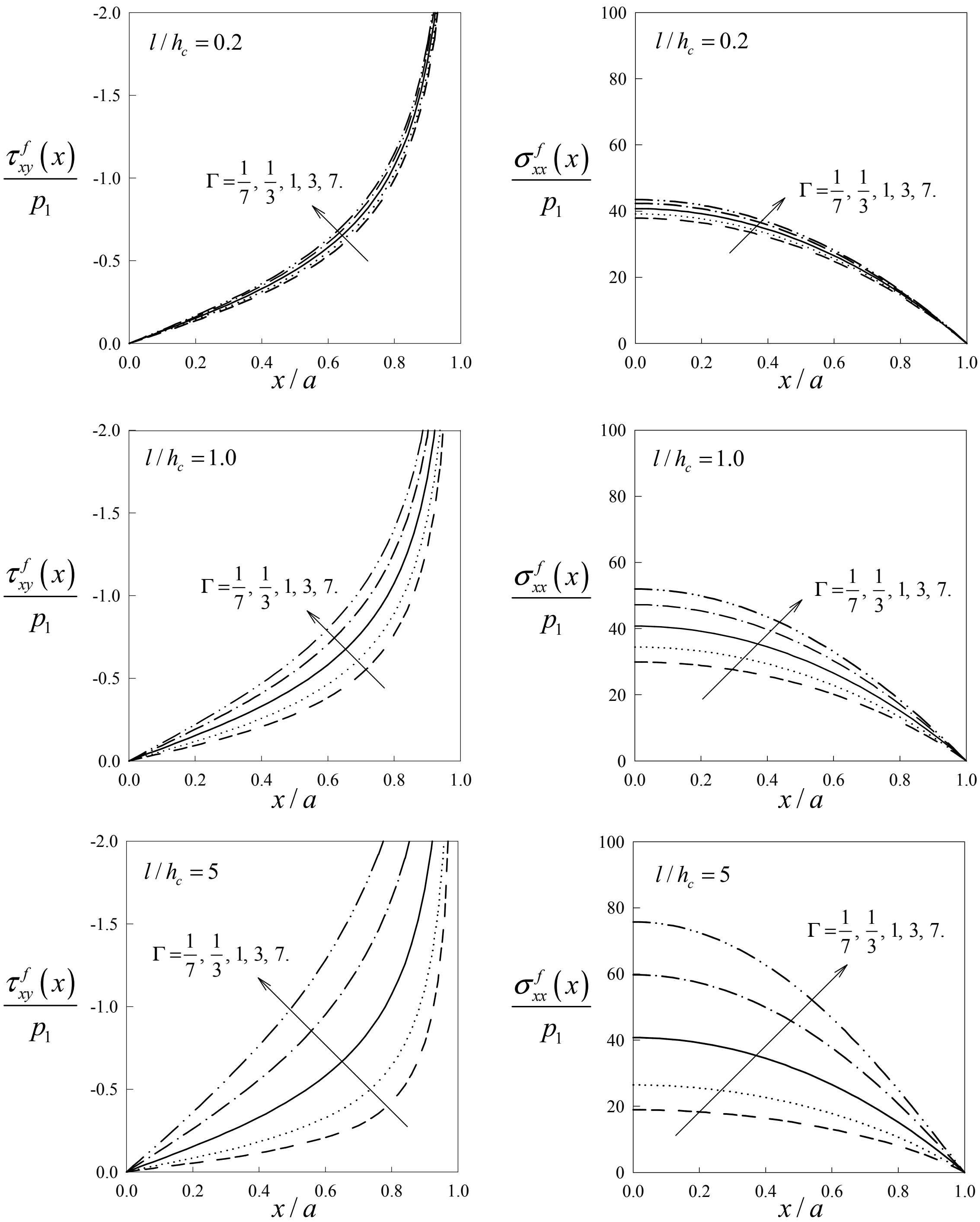

Figure 2 Interfacial shear stress $\tau_{xy}^f(x)$ and the axial normal stress $\sigma_{xx}^f(x)$ in the film for various values of the nonhomogeneity parameter $\Gamma = \mu_s / \mu_0$ and for various values of the parameters l/h_c with $\nu_c = \nu_f = 0.3$, $\lambda = \mu_0 / \mu_f = 0.01$, $l/h_f = 100$. Note that arrows given in the figures shows the increasing direction of the variant parameters thereof.

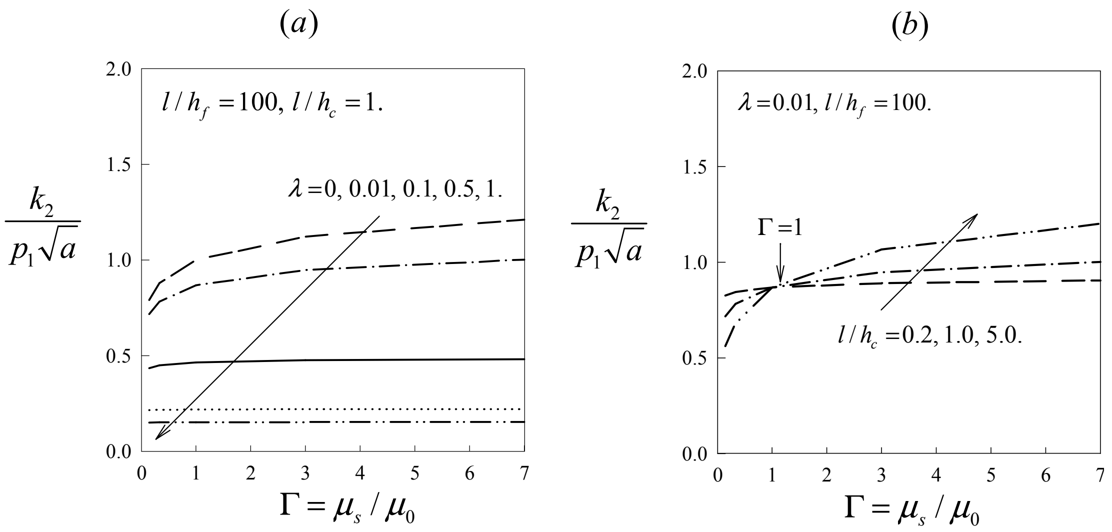

Figure 3 Mode II stres intensity factor at the film edge for various values of the parameters $\Gamma = \mu_s / \mu_0$, $\lambda = \mu_0 / \mu_f$ and l / h_c Note that arrows given in the figures shows the increasing direction of the variant parameters thereof.

Conclusions

In the light of the figures the following conclusions can be drawn:

- As the film length to coating thickness ratio, l / h_c, increases, the normal stress $\sigma_{xx}^f(x)$ in the film increases, and the grading gets more effective to increase or decrease the stress values (see Fig. 2).
- For fixed values of l / h_c, and l / h_f, as the film gets stiffer than the underneath material, mode II stress intensity factor, k_2, increases and the effect of the value of the grading ratio, Γ, becomes more crucial (see Fig. 3a).
- For fixed values of l / h_f, λ, and Γ, any change in the film length to coating thickness ratio, l / h_c, has no effect on k_2 value. When grading ratio becomes less than unity, $\Gamma < 1$, for a fixed film length, a thinner coating; and when grading ratio becomes greater than unity, $\Gamma > 1$, a thicker coating should be selected to decrease the stress intensity factor at the film ends (see Fig. 3b).

The results of this study may be used as a starting point for a designing the thin films bonded to graded materials as long as the hypotheses of this study is valid enough for the related applications.

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